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Module 2 Homework

coach
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1. Researchers are interested in estimating the natural log of the proportion of people in the population with hypertension. In a random sample of n subjects, let X be the number with hypertension. **1 / 1 point**

Create a confidence interval for the natural log of the proportion of people with hypertension. Assume that n is very large.

- (selected) $$\log(\hat{p}) \pm Z_{1-\alpha/2}\sqrt{\frac{1-\hat{p}}{\hat{p}n}}$$
- The answer cannot be determined from the information given.
- $$\log\left(\hat{p} \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right)$$
  where $\log$ acts componentwise
- $$\log(\hat{p}) \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

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Correct
$f(x) = \log(x)$. Then $f'(x) = 1/x$. Also $Var(\hat{p}) = p(1-p)/n$. Then the standard error is

$$\frac{1}{p}\sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{1-p}{pn}}.$$

Thus, the confidence interval is

$$\log(\hat{p}) \pm Z_{1-\alpha/2}\sqrt{\frac{1-\hat{p}}{\hat{p}n}}$$

2. If the odds are 4 to 1 (i.e., just 4) of the NFL Ravens losing, what is the implied probability that they win? **1 / 1 point**

- 0.80
- 0.40
- 0.30
- 0.70
- 0.10
- 0.60
- 0.50
- (selected) 0.20

Correct
We have that

$$\frac{1-p}{p} = 4$$

where p is the probability that the Ravens win. Then $p = 1/5$ and $1 - p = 4/5$.

3. In a randomly sampled survey of self-reported stress levels from two occupations, the following data were obtained where there were 100 in each group **1 / 1 point**

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	High Stress	Low Stress
Professor	70	
Lion Tamer	15	

What is the value of the score Z test statistic for evaluating equality of the two proportions of high stress?

- (selected) Around 8 (or -8, depending on which you subtract first)
- Around 5 (or -5, depending on which you subtract first)
- Around 4 (or -4, depending on which you subtract first)
- Around 3 (or -3, depending on which you subtract first)

Correct
$H_0 : p_{Prof} = p_{LT}$ versus $H_a : p_{Prof} \neq p_{LT}$

```
p1 <- 0.7
p2 <- 0.15
p <- 85/200
ts <- (p1 - p2)/sqrt(p * (1 - p) * 2/100)
ts
```

```
[1] 7.867
```

4. Consider the following data recording case status relative to an environmental exposure **1 / 1 point**

	Case	Control
Exposed	45	21
Unexposed	15	52

What would be the estimated asymptotic standard error for the log odds ratio for this data?

- Around 0.2
- (selected) Around 0.4
- Around 1.5
- Around 1.2

Correct

```
sqrt(1/45 + 1/15 + 1/21 + 1/52)
```

```
[1] 0.3946
```

5. If $x \sim Binomial(n, p)$ and a $Beta(2, 2)$ prior is placed on p, what is the posterior mean for p? **1 / 1 point**

- (selected) $$\frac{x+2}{n+4}$$
- $$\frac{x}{n}$$
- $$\frac{x+2}{n+2}$$
- $$\frac{x+1}{n+1}$$

Correct

$$p^x(1-p)^{n-x} \times p(1-p) = p^{x+2-1}(1-p)^{n-x+2-1}$$

Thus p given the data is Beta$(x+2, n-x+2)$. This has mean

$$\frac{x+2}{n+4}$$

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6. Researchers conducted a blind taste test of Coke versus Pepsi. Each of four people was asked which of two blinded drinks given in random order they preferred. The data was such that 3 of the 4 people chose Coke. Assuming that this sample is representative, report a P-value for a test of the hypothesis that Coke is preferred to Pepsi using a one-sided exact test. **1 / 1 point**

- Around 0.4
- Around 0.2
- Around 0.1
- (selected) Around 0.3

Correct
Let p be the proportion of people who prefer Coke. Then, we want to test $H_0 : p = .5$ versus $H_a : p > .5$. Let X be the number out of 4 that prefer Coke; assume $X \sim Binomial(p, .5)$.
$Pvalue = P(X \geq 3) = \text{choose}(4,3)0.5^3 0.5^1 + \text{choose}(4,4)0.5^4 0.5^0$

```
pbinom(2, size = 4, prob = 0.5, lower.tail = FALSE)
```

```
[1] 0.3125
```

```
choose(4, 3) * 0.5^4 + choose(4, 4) * 0.5^4
```

```
[1] 0.3125
```